

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10	(a)			$[H^+] = 10^{-3.24}$ (1) $[H^+] = 5.75 \times 10^{-4} \text{ mol dm}^{-3}$ (1)		2		2	2	
	(b)	(i)		filtration		1		1		1
		(ii)	I	excess sodium hydroxide dissolves any precipitate produced by zinc (1) excess ensures that all iron ions are precipitated (1)			2	2		2
			II	moles $\text{Fe}_2\text{O}_3 = \frac{14.5 \times 10^{-3}}{159.6} = 9.085 \times 10^{-5} \text{ mol}$ (1) concentration = $\frac{2 \times (9.085 \times 10^{-5})}{0.025} = 7.27 \times 10^{-3} \text{ mol dm}^{-3}$ (1)		2		2	2	
			III	$\frac{0.2}{14.5} \times 100 = 1.4\%$		1		1	1	
		(iii)	I	solution turns (pale) pink do not accept purple to pink	1			1		1
			II	moles of $\text{MnO}_4^- = 3.80 \times 10^{-5} \text{ mol}$ (1) $[\text{Fe}^{2+}] = 7.60 \times 10^{-3} \text{ mol dm}^{-3}$ (1) ecf possible		2		2	2	